Predicting the American Dream: Do Wages Reliably Forecast Housing Prices?

Group 5
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Project Details

Motivation:

- Research has determined that there is an association between housing prices and health outcomes, in that being able to afford housing leads to better physical and mental health.
- Being young adults who will be searching for homes in the near future, we found it important to better understand the factors that drive housing affordability, particularly whether changes in wages can help predict shifts in housing prices and inform more accessible pathways to homeownership.

Hypotheses:

- Average hourly wages in the United States Granger-cause the U.S. National Home Price Index (past wage values improve prediction of housing prices).

Research Question:

- Is average hourly wage in the United States a reliable predictor of the U.S. National Home Price Index?

Modeling Approach:

- Granger Causality Analysis

Goal:

- To determine if average hourly wage data in the United States can forecast activity of the U.S. National Home Price Index.

Data Acquisition & Explanation

Column	Description	Potential Responses
date	Date of observation	3/1/2006 - 12/1/2025
date_diff	Number of days over which wage and USNHPI were measured	28 days, 30 days, 31 days
avg_hourly_wage	Average hourly wage in the U.S. in given month	20.04, 33.02, 37.02
avg_hourly_wage_lag	Lagged average hourly wage to be used as predictor	20.04, 33.02, 37.02
avg_hourly_wage_diff	Shows the change in average hourly wage from previous month	0, -0.04, 0.11
avg_hourly_wage_log_diff	The log transform of average hourly wage difference	0.004460973940624946, -0.001985112314420334
USNHPI	Raw index value for each month	182.748, 141.037, 327.455
USNHPI_diff	Shows change in U.S. National Home Price Index from previous month	0.9, -1.112, 4.998
USNHPI_log_diff	The log transform of USNHPI difference	0.003977963444496524, -0.0011002383152698414

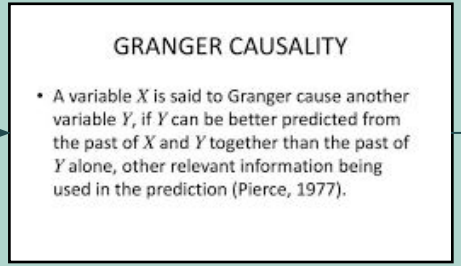
Acquisition:
The original two datasets were acquired from the Federal Reserve Bank of St. Louis website with no licensing or ethical concerns. The files were both in a CSV format with ~240 rows and 2 columns.

We chose this option due to the large time period that this data allowed us to cover (~20 years).

We then cleaned our data set by standardizing responses and correcting variable types. We also created new variables by transforming those from the original data.

This brought us to our final data set that would be utilized in our analysis (seen to the left).

Analysis Plan & Justification



- Clean data
- Standardize responses
- Ensure variable types are correct

- Test stationarity of data
- Choose lag order
- Run analysis to determine predictive power

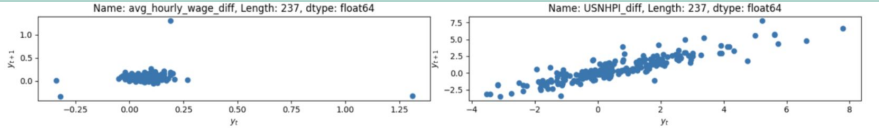
- Determining the statistical significance of the predicted power or wages compared to USNHPI

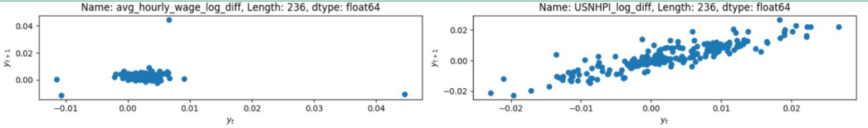
Challenges: the data was not completely stationary, additional steps had to be taken to determine how to proceed with Granger Causality analysis

Diagnostic/validation work:

- lag plots, KDSS, and ADF tests were used to test stationarity of the 2 variables of interest
- Lag order p in Var(p) model was selected based on 4 different multivariate information criterion
- Granger-causality was determined by SSR-based chi-squared test

Tricky Analysis Decision





kpss test results:

	avg_hourly_wage_diff	USNHPI_diff
Test Statistic	2.276494	1.040458
P-value	0.010000	0.010000
Critical Value 1%	0.739000	0.739000
Critical Value 2.5%	0.574000	0.574000
Critical Value 5%	0.463000	0.463000
Critical Value 10%	0.347000	0.347000

adf test results:

	avg_hourly_wage_diff	USNHPI_diff
Test Statistic	-2.354742	-2.170029
P-value	0.154919	0.217303
Critical Value 1%	-3.459490	-3.460019
Critical Value 5%	-2.874358	-2.874590
Critical Value 10%	-2.573602	-2.573725

kpss test results:

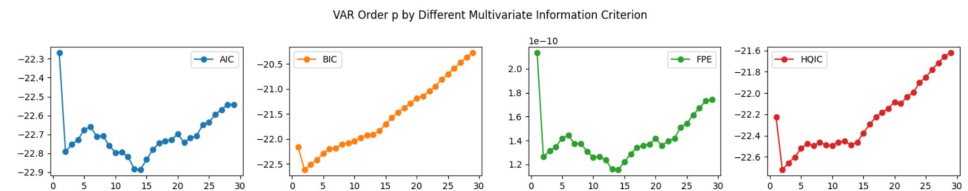
	avg_hourly_wage_log_diff	USNHPI_log_diff
Test Statistic	1.103137	0.970223
P-value	0.010000	0.010000
Critical Value 1%	0.739000	0.739000
Critical Value 2.5%	0.574000	0.574000
Critical Value 5%	0.463000	0.463000
Critical Value 10%	0.347000	0.347000

adf test results:

	avg_hourly_wage_log_diff	USNHPI_log_diff
Test Statistic	-1.331651e+01	-2.295389
P-value	6.586718e-25	0.173460
Critical Value 1%	-3.458608e+00	-3.460154
Critical Value 5%	-2.873972e+00	-2.874649
Critical Value 10%	-2.573396e+00	-2.573757

- Granger Causality Test requires stationarity
- avg_hourly_wage_diff and USNHPI_diff are the first differences
  - Results from the KPSS and ADF tests show non-stationarity
- After a logarithmic transformation of avg_hourly_wage and USNHPI and taking the first difference, we get avg_hourly_wage_log_diff and USNHPI_log_diff
  - avg_hourly_wage_log_diff can now be considered stationary, USNHPI_log_diff is still non-stationary
- We decided to work with the avg_hourly_wage_log_diff and USNHPI_log_diff because this is standard in macroeconomic data and taking second differences may remove too much information to answer our research question

Bias & Uncertainty Validation




```
AIC     14
BIC      2
FPE     14
HQIC     2
dtype: int64
```

Biases:

- Housing prices and wages vary greatly by region, meaning results may not apply equally to all local housing markets
- Major economic events like the 2008 financial crisis and COVID-19, events that affected both wages and housing prices and could influence the relationship we are studying
- Left out other data like interest rates, so wage-price links might be shared trends, instead of wages “causing” housing prices

Biases Corrected For:

- We report Granger results as in-sample predictability and do not treat wage-price patterns that might actually come from shared macro conditions as a settled, causal story
- Conclusions are for one long national monthly history, not a world where the economy behaves the same every year

Uncertainty Validation:

- To characterize uncertainty in our Granger results, we use the p-values from the chi-squared tests in our script, with $\alpha = 0.05$, fitted on the first 80% of months (189 training rows after preprocessing). They are in-sample measures of statistical evidence
- We report results at both $p = 2$ and $p = 14$ as the plot above shows the choice of p is not stable across standard rules
- Stationarity checks (lag plots, KPSS, ADF) in output; still reliant on log differences for the main tests

Results & Conclusions

The lag order p defines the maximum number of past time periods of avg_hourly_wage_log_diff included as predictors for USNHPI_log_diff.

```
Granger Causation Matrix for p=2:
                              avg_hourly_wage_log_diff_x  USNHPI_log_diff_x
avg_hourly_wage_log_diff_y                     1.000000           0.212141
USNHPI_log_diff_y                              0.429672           1.000000
```

No Granger-causality detected when p=2.

```
Granger Causation Matrix for p=14:
                              avg_hourly_wage_log_diff_x  USNHPI_log_diff_x
avg_hourly_wage_log_diff_y                     1.000000           0.212141
USNHPI_log_diff_y                              0.023672           1.000000
```

Granger-causality detected when p=14.

We conclude that avg_hourly_wage_log_diff Granger-causes USNHPI_log_diff only at high lag orders (p=14). There is no evidence of Granger-causality at p=2.

Next Steps

New Lines of Exploration

Accounting for external “shocks” (e.g. 2008 crisis) to better understand how relationships evolve over time

Include more possible predictors (multivariate VAR)

Project Improvements

Adding macroeconomic variables such as interest rates (mortgage rates) and inflation (CPI)

Collect more data to improve stationarity results

New Questions

Can wage trends realistically be used to anticipate housing affordability challenges?

Does the predictive relationship vary across regions or differing income levels?

References

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